CIE AS PHYSICS (9702) • QUESTION PAPER

7.2 Transverse & Longitudinal Waves

Syllabus Section: 7. Waves **Paper Type:** Multiple Choice (Paper 1) **Total:** 14 Questions (14 Marks)

Question 1 Easy [1 mark]

The following statements are about waves.

1	Waves can travel through free space
2	Waves can be polarised
3	Waves have a frequency inversely proportional to their wavelength
4	Waves have an intensity proportional to their amplitude

Which ones would apply to both sound and light waves?

A 3 only

B 4 only

C 1 and 4

D 1 and 2

Question 2 Easy [1 mark]

Ultraviolet waves have an order of magnitude of the frequency of the longest-wavelength of 10^{x} Hz.

What is the value of x ?

A 19

B 17

C 16

D 13

Question 3 Easy [1 mark]

Which of the following statements shows that sound waves are longitudinal?

A sound is diffracted around corners

B sound is refracted as it passes from hot air to cold air

C sound cannot be polarised

D sound can be reflected from a solid surface

Question 4 Easy [1 mark]

Fig 1.1 shows a longitudinal wave.

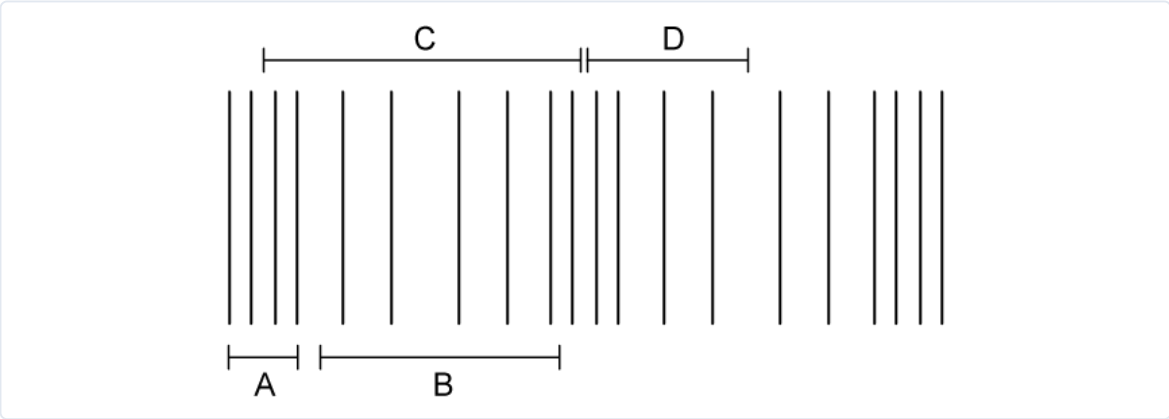


Fig. 1.1

Which label correctly shows the wavelength of a longitudinal wave?

- **A**
- **B**
- **C**
- **D**

Question 5 Easy [1 mark]

Which pair contains one transverse and one longitudinal wave?

A.	Water wave	Visible light
B.	X-ray	Ultrasound
C.	Primary (P) wave	Sound wave
D.	Infrared wave	Vibration on a guitar string

A

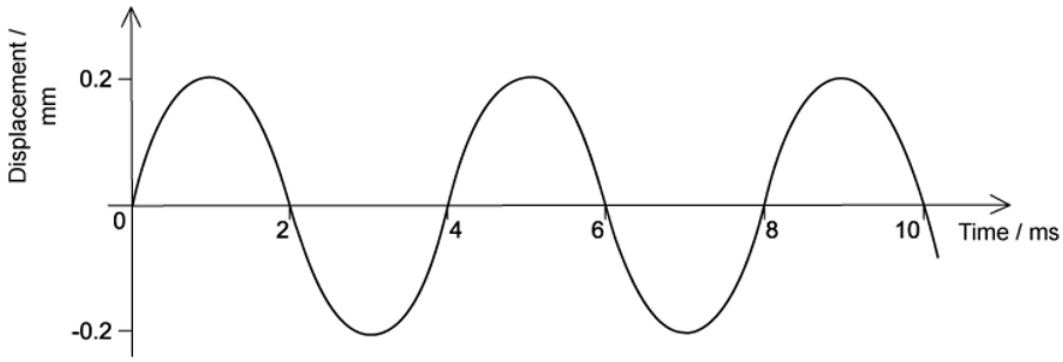
B

C

D

Question 6 Medium [1 mark]

Sound travels with a speed of 330 m s^{-1} in air. The variation with time of the displacement of an air particle due to a sound wave is shown below.



Which of the following statements about the wave is correct?

A the wavelength of the sound wave is 1.32 m

B the graph shows that the sound is a transverse wave

C the frequency of the wave is 500 Hz

D the intensity of the wave will be doubled if its amplitude is increased to 0.4 mm

Question 7 Medium [1 mark]

A wave is travelling through a medium at 24 m s^{-1} with a frequency of 15 Hz.

What is the phase difference between two points 2 m apart?

A They are out of phase by half a cycle.

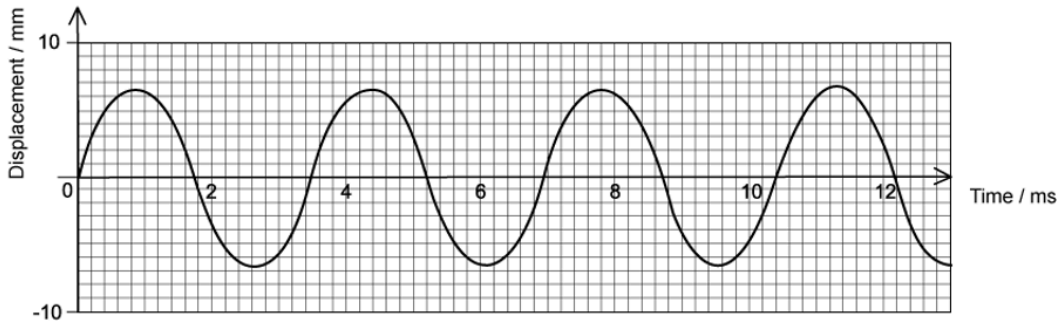
B There is no phase difference.

C They are out of phase by a quarter of a cycle.

D They are out of phase by 0.8 of a cycle.

Question 8 Medium [1 mark]

What is the frequency, amplitude and period of the wave shown in the graph below, to 2 significant figures?



	Period / s	Frequency / Hz	Amplitude / m
A	0.0042	240	0.013
B	0.0035	290	0.0065
C	0.0031	320	0.013
D	0.0027	290	0.0065

A

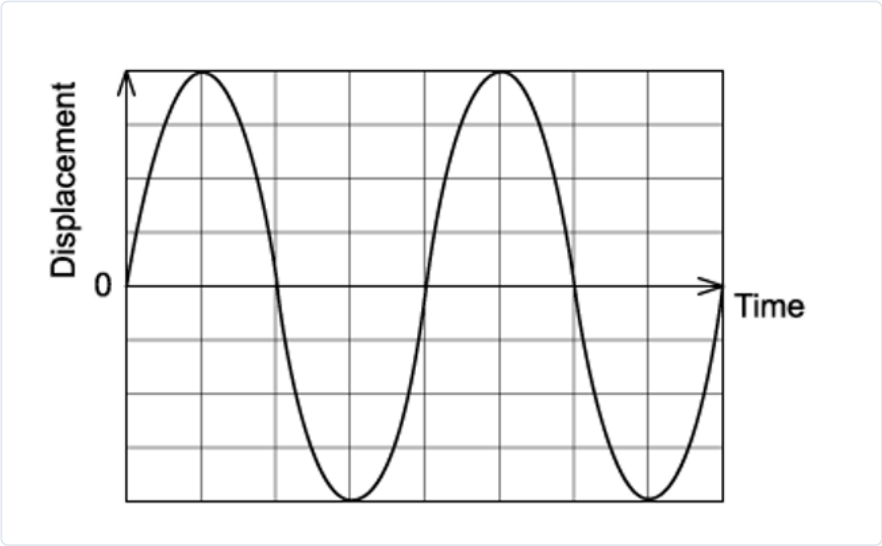
B

C

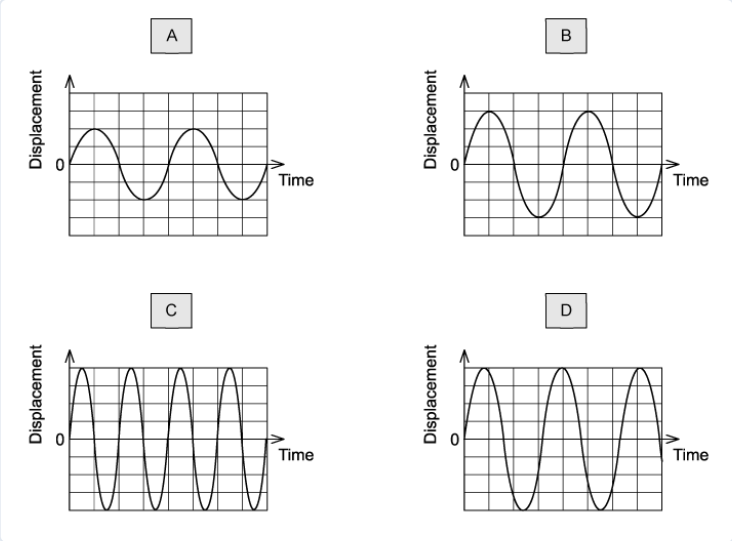
D

Question 9 Medium [1 mark]

The graph shows the displacement of a sound wave over time.



When the intensity of the wave is halved, which graph shows the correct displacement of the wave ?



A

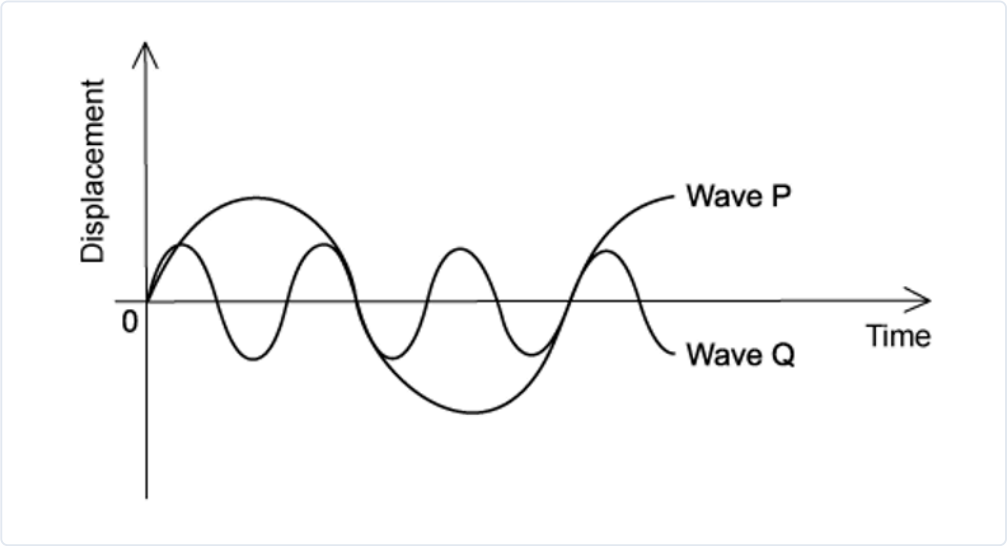
B

C

D

Question 10 Hard [1 mark]

The graph shows two waves P and Q.



Wave P has an amplitude of 8 cm and a frequency 100 Hz.

What is the amplitude and the frequency of wave Q?

	Amplitude / cm	Frequency / Hz
A	4	300
B	4	33
C	2	300
D	2	33

A

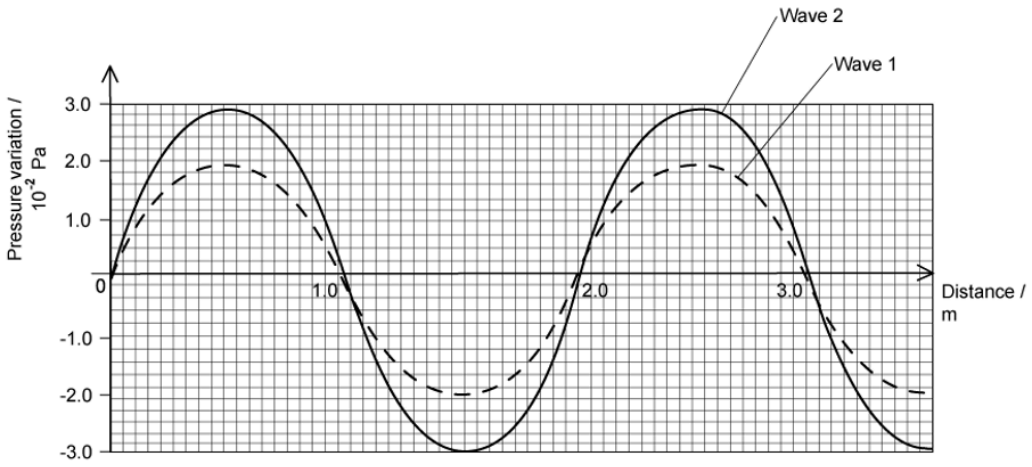
B

C

D

Question 11 Hard [1 mark]

The graph below shows two sound waves, the sound wave is a series of moving pressure variations.



Wave 1 has an intensity of 1.6×10^{-6} W m^{-2}, what is the intensity of wave 2 ?

A 2.4×10^{-6} W m^{-2}

B 3.0×10^{-6} W m^{-2}

C 3.6×10^{-6} W m^{-2}

D 4.5×10^{-6} W m^{-2}

Question 12 Hard [1 mark]

Fig 1.1 shows a displacement-distance graph of a transverse wave at t = 0 s. The wave is travelling at 64 m s^{-1}. Particle R is marked at a position of 0.2 m.

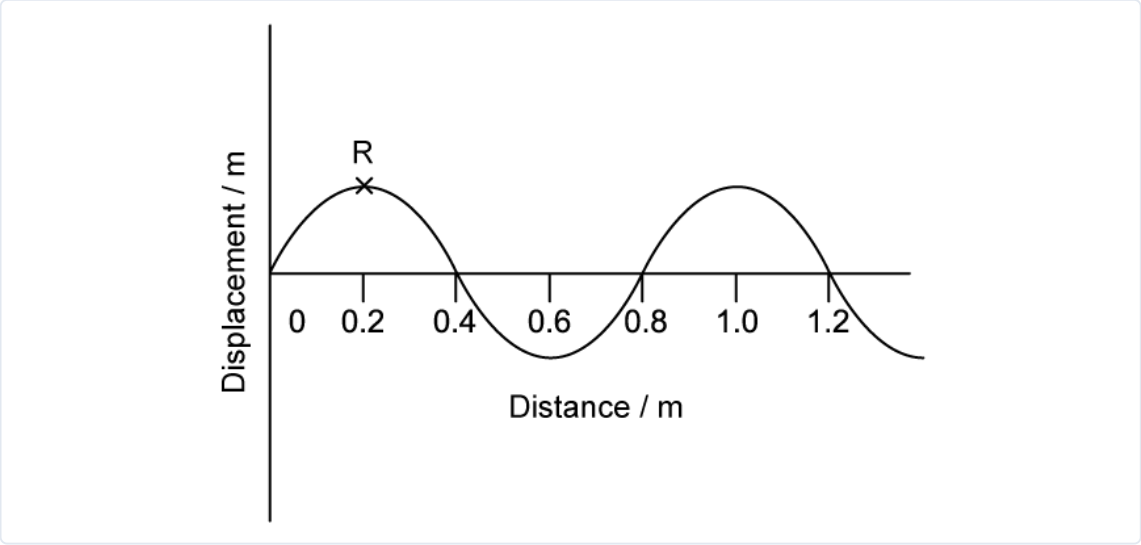


Fig. 1.1

What would be the phase difference between the position of Particle R at t = 0 s and t = 0.034 s?

A 7.0°

B 49°

C 100°

D 260°

Question 13 Hard [1 mark]

A source emits a sound wave and moves away from an observer at a speed of 85 m s^{-1}. Fig 1.1 shows the displacement-distance graph of the sound wave at the source.

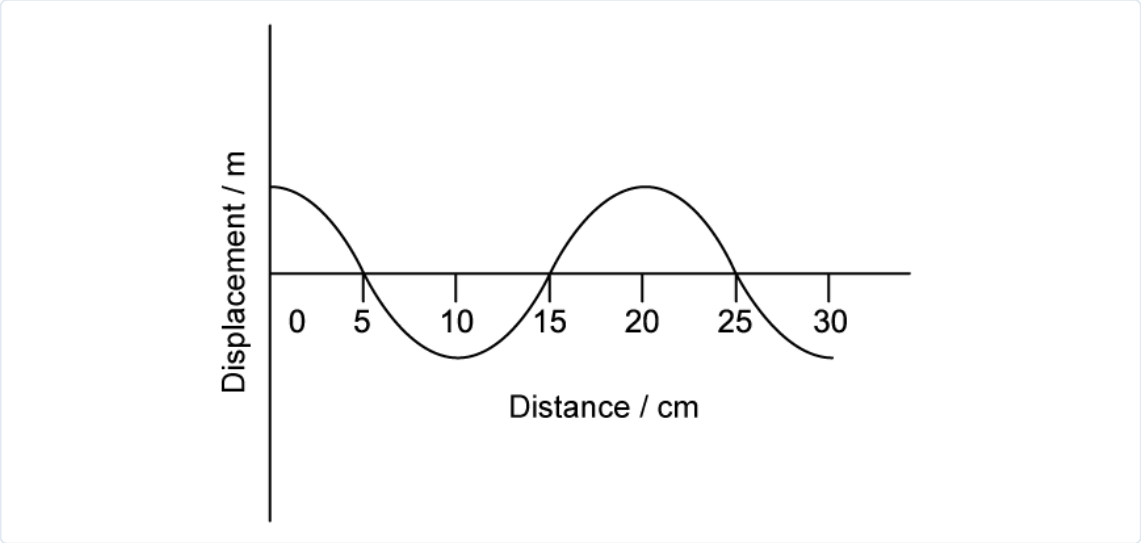


Fig. 1.1

What is the observed frequency of the sound wave?

The speed of sound, v = 340 ms^{-1}

A 1300 Hz

B 1400 Hz

C 1700 Hz

D 2100 Hz

Question 14 Hard [1 mark]

Fig 1.1 shows two longitudinal waves, wave A and wave B, over time *t*. Wave A has an intensity of I_A.

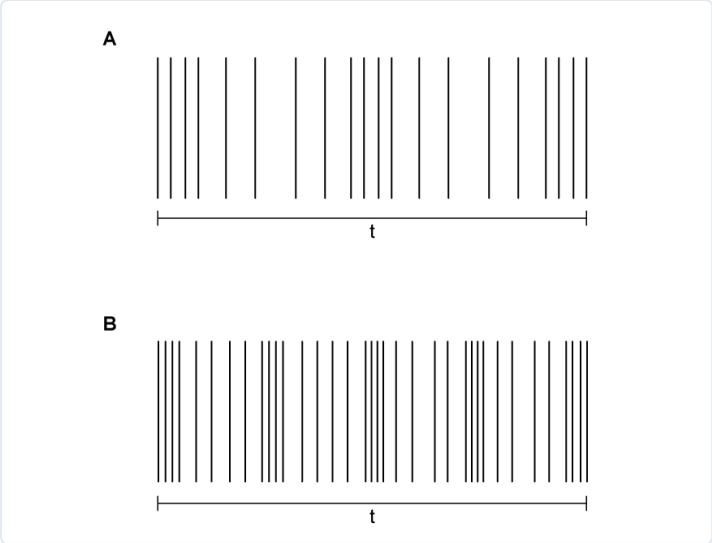


Fig. 1.1

Assuming both waves have equal amplitude, what is the intensity of wave B?

A $\frac{1}{4}I_A$

B $\frac{1}{2}I_A$

C $2I_A$

D $4I_A$